AGU Advances

Supporting Information for

Interpreting changes in global methane budget in a chemistry-climate model constrained with methane and isotopic observations

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Introduction

The Supporting Information includes detailed model evaluations with surface and satellite observations, additional discussions on emission sensitivity, as well as additional information on model inputs.

Text S1. Model evaluations with surface observations

We use measurements of CH_4 and $\delta^{13}CH_4$ from a globally distributed network of air sampling sites maintained by Global Monitoring Laboratory (GML) (Lan et al., 2024; Michel et al., 2023) of the National Oceanic and Atmospheric Administration (NOAA) to evaluate model performance. Our model with optimized methane emissions is able to capture the global trend of methane (Lan et al., 2025) and $\delta^{13}CH_4$ (Michel et al., 2023). Despite optimizing emissions uniformly at a global scale instead of regionally as described in the main text, our model is still able to capture the latitudinal distributions of methane. Figure S1 shows the comparison of simulated surface CH_4 concentrations and $\delta^{13}CH_4$ with NOAA GML surface observations over multiple sites. Since the observations represent well-mixed background conditions, there is no significant difference among different experiments at a global scale (Figure 4 in main text). However, there are noticeable differences between EXP1 and EXP2 over different latitude bands (Figures 3&4 in main text). Here we only show the evaluation of EXP1 and EXP2 in Figure S1 over several NOAA GML sites.

Both EXP1 and EXP2 generally capture the monthly variabilities of methane concentrations across different sites, with correlation coefficient (R) close to 1. Also, the model is able to reproduce methane growth rates, with R > 0.5 over most of the sites. With different source contributions, there are overall better estimates of $\delta^{13}CH_4$, especially at the Mt. Waliguan site (WLG), China, where EXP1 shows much better monthly variation. Despite this being a remote site, previous studies have shown methane concentrations can be affected by the long-range transport from source regions (Liu et al., 2021). The higher fossil fuel emissions over China in EXP1 (blue line) would lead to less negative total $\delta^{13}CH_4$ (also closer to the observed $\delta^{13}CH_4$) than that from EXP2 (red line) at a regional (country-wide) scale. In summary, EXP1 generally shows better model performance than EXP2.

Text S2. Model evaluations with satellite observations

We also compare the model simulated column average concentration of CH_4 (XCH_4) with the retrieved XCH_4 from the Greenhouse gases Observing SATellite (GOSAT) (Yoshida et al., 2013). We sample the model profiles at specific locations with applying averaging kernels to get model simulated XCH_4. Due to the limited computing resources, we have model output on a monthly basis. Therefore, we extract the monthly mean model profiles at specific locations scanned by the GOSAT. This would lead to uncertainty when comparing the magnitudes of XCH_4, but still provide some information on model performance of spatially distributed XCH_4.

Figure S2 shows the spatial and temporal comparison of simulated and observed XCH_4. Despite the different source contributions, the simulated XCH_4 are pretty comparable across all the experiments at both global and regional scales. In general, there is a consistent low bias (about 23 ppb) across all the four experiments, which is due in part to the different sampling frequency of simulated methane profiles. However, there is a pretty good spatial correlation between model estimates and satellite observations (with R > 0.9). In addition, all experiments show good temporal correlation (monthly variability) with $R^2 > 0.9$. Similarly, EXP2 shows a larger model bias over the Northern Hemisphere continents, especially over East Asia (Figure S2f) than EXP1, suggesting better ENE estimates from CEDS than EDGAR (particularly over China).

The evaluation suggests the reasonable spatial and temporal representations of methane emissions. Despite the differences in the sectoral sources among four experiments, there is not much difference in the simulated methane concentrations from four experiments.

Text S3. Emission sensitivity to different factors

There are four experiments conducted in the model to explore the sensitivity of emissions to different inputs and isotopic fractionations. As shown in Figure S3, The difference in the fossil fuel emissions (energy sector, ENE) between CEDS (Hoesly et al., 2018) (EXP1) and EDGARv5.0 (Monforti-Ferrario et al., 2019) (EXP2) leads to the different optimized wetland (WET) and biomass burning (BMB) emissions. Since ENE emissions are lower in EDGAR than CEDS (by about 25±12 Tg yr^{-1}), we need to have higher total of WET and BMB to make sure the total methane emissions are same between EXP1 and EXP2, which are optimized (constrained) by the observed methane concentrations. Further constrained by the observed atmospheric $\delta^{13}CH_4$, WET emissions are about 13±6 Tg yr^{-1} higher and BMB emissions are about 12±7 Tg yr^{-1} higher in EXP2. If we simply increase isotopic fractionation (EXP3) without changing the emission totals, we need to reduce WET emissions in EXP1 by about 9±2 Tg yr^{-1} and put this amount to BMB emissions to force the model capture the observed atmospheric $\delta^{13}CH_4$ trend. We also tested the model forced by MERRA2 meteorology (EXP4). Compared to the estimates forced by NCEP meteorology (EXP1), the total methane emissions have to be higher by 11±4 Tg

yr^{-1}, with 9±2 Tg yr^{-1} higher in WET emissions and 2±1 Tg yr^{-1} higher in BMB emissions, constrained by the CH_4 and $\delta^{13}CH_4$ observations.

As global total methane emissions have increased since 2006, the contribution of AGR &WST to the total methane emissions do not change much from 1999-2006 to 2007-2017 (about 36%), while the contribution of ENE to the total methane emissions increases from 21.1% in 1999-2006 to 24.6% in 2007-2017 following CEDS ENE emissions (EXP1) and from 15.5% to 18.3% following EDGAR v5.0 ENE emissions (EXP2). The overall decrease in the global total wetland emissions are minor (< 4%), however, with higher total emissions during 2007-2017, the contributions from wetlands tend to decrease. With a moderate decrease in biomass burning emissions during 2007-2017, there is also a decrease in its contribution to total methane emissions.

The optimized total methane emissions are sensitive to the simulated hydroxyl radical (OH), which has been discussed in our previous studies (He et al., 2020; He et al., 2021). The optimized emissions for specific sectors are also sensitive to the initial emission inventory and isotopic fractionation. In this work, we assume the emission trends of other sources sectors (e.g., agriculture & waste) are reasonably represented in the CEDS inventory. We test the different trends for ENE, which leads to about 12-13 Tg yr^{-1} difference in WET and BMB emissions. It is also possible, with different trends and magnitudes of the agricultural (AGR) emissions, there would also be noticeable differences in WET and BMB emissions. In this work, we add 25 Tg yr^{-1} emissions to CEDS AGR emissions each year (keep AGR emission trend unchanged) to bring it closer to that in EDGAR v5.0. The reason we add this amount is to make sure when we perturb other factors, we can still have reasonable estimates in WET and BMB emissions (at least positive emissions).

In theory, there are many ways to perturb different source sectors while forcing the model to reproduce the observed trends of CH_4 and $\delta^{13}CH_4$, which would lead to different solutions. The purpose of testing different fossil fuel emissions from two inventories is to investigate the fossil fuel contribution to the methane growth, rather than demonstrate which inventory is better. Also, we optimize WET and BMB sectors due to their large interannual variability and uncertainty in the emission estimates. Nevertheless, based on all the sensitivity simulations, compared to the relatively stable methane of 1999-2006, the post-2006 renewed methane increase is more likely due to the emission increase from ENE and AGR sectors that are large enough to offset the increases in sinks.

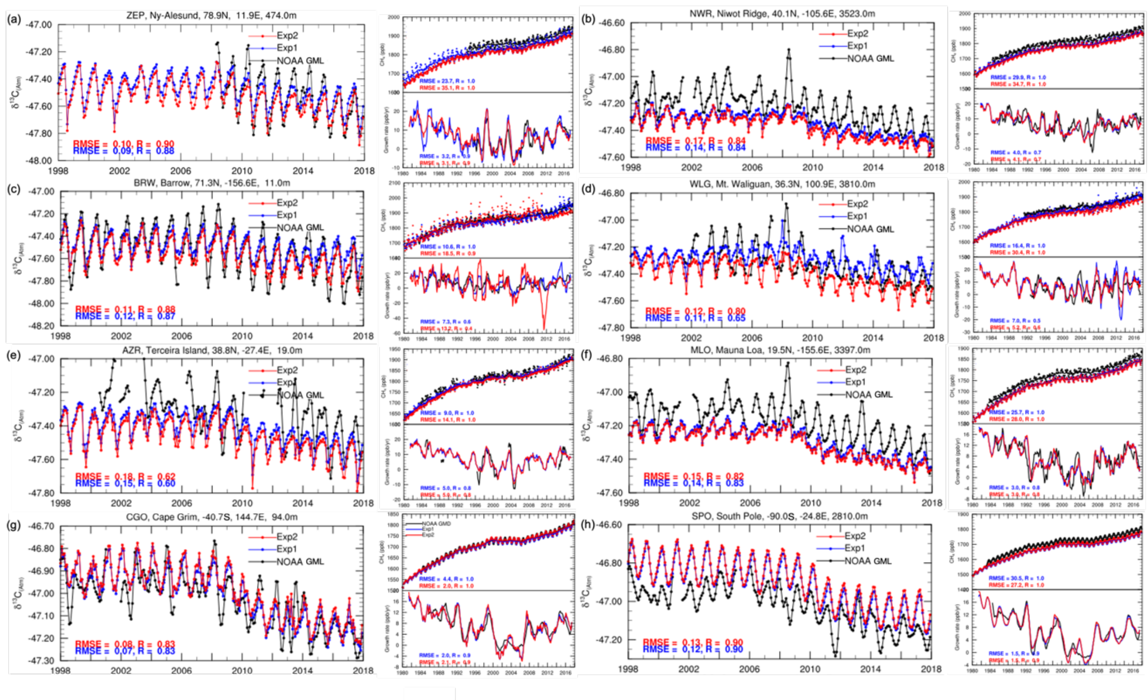

Figure S1. Time series of simulated monthly mean CH_4 and $\delta^{13}CH_4$ (blue lines for EXP1 and red lines for EXP2) and observations from selected NOAA GML sites (black lines). Each panel (a-h) includes three figures, with the left column for monthly $\delta^{13}CH_4$ (site information is shown above the figure) and right column for monthly methane concentrations (top) and methane growth rate (bottom).

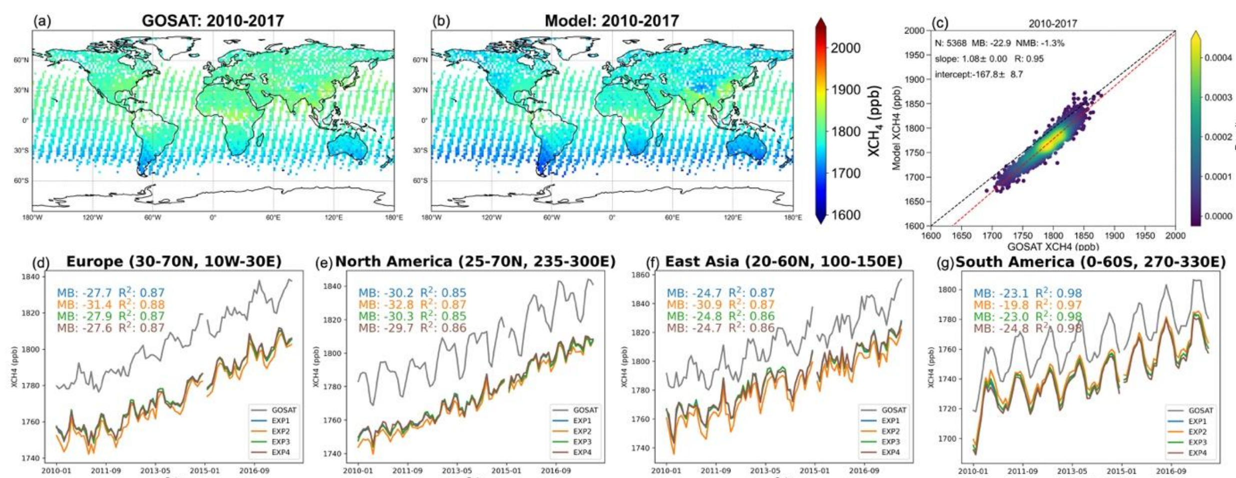

Figure S2. Comparison of simulated column averaged methane concentrations (XCH_4) with GOSAT observations. Spatial distributions of 2010-2017 multi-year mean XCH_4 from GOSAT and EXP1 are shown panel (a) & (b). Statistics for spatial comparisons are calculated for EXP1 (c). Statistics for temporal correlations for four experiments for different regions are shown in panel d-g (EXP1 in blue, EXP2 in orange, EXP3 in green, and EXP4 in brown).

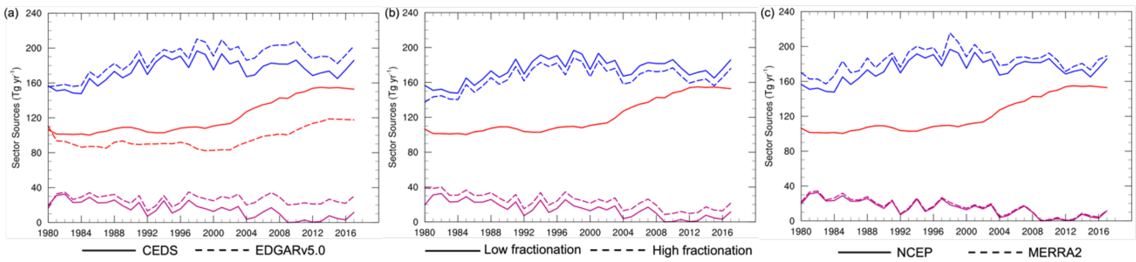

Figure S3. Emission sensitivities to different initial emissions (a), OH fractionations (b), and meteorological inputs (c). Blue lines represent wetland emissions, red lines represent fossil fuel emissions, and purple lines represent biomass burning emissions. Solid and dashed lines represent emissions under different experiment scenarios.

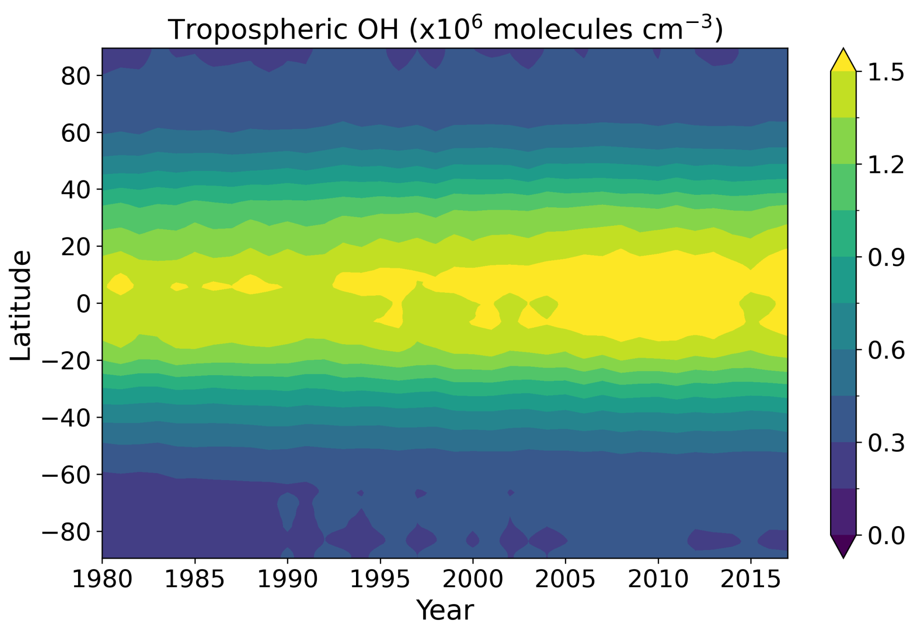

Figure S4. Time series of latitudinal distribution of annual mean air-mass weighted tropospheric OH concentrations.

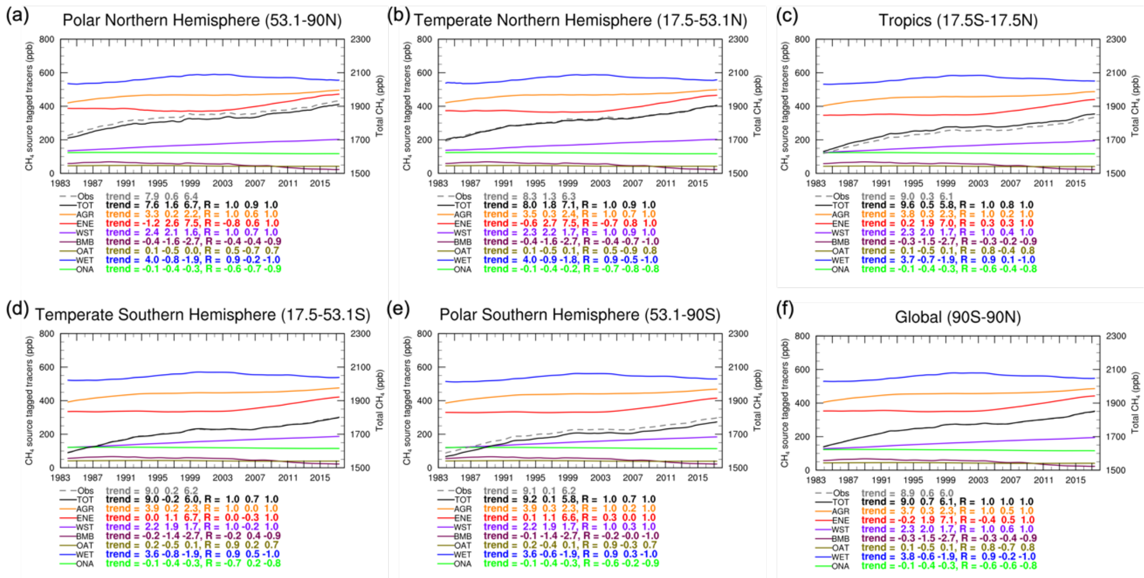

Figure S5. Linear trend of source tagged tracers (described in main text) across different latitude bands. The calculated trend and correlation with the observed trend are listed below the figure. Gary dashed lines represent observed trend calculated from GML MBL sites and solid lines represent trend calculated from each source tagged tracers.

Table S1. Methane Emission Inventories.

Source Category	Database	Temporal Variability	References
Anthropogenic*	CEDS v2017-05-18	1980-2014 monthly	Hoesly et al. (2018)
	SSP2-4.5	2015-2017 monthly	Gidden et al. (2018)
Biomass Burning	BB4MIP	1980-2014 monthly	van Marle et al. (2017)
	SSP2-4.5	2015-2017 monthly	Gidden et al. (2018)
Wetlands	WetChart v1.0	2001-2015 monthly Climatological monthly mean for rest of years	Bloom et al. (2017)
Ocean	MOZART	Climatological monthly mean	Brasseur et al. (1998)
Near-shore	TransCom-CH4	Climatological annual mean	Lambert and Schmidt (1993), Patra et al. (2011)
Termites	NASA-GISS	Climatological annual mean	Fung et al. (1991)
Mud volcanoes	TransCom-CH4	Climatological annual mean	Etiope and Milkov (2004), Patra et al. (2011)

*Anthropogenic sectors: Agriculture, Energy, Industry, Transport, Residents, Shipping, and Waste

Table S2. Global mean isotopic ratios used to generate spatially resolved isotopic ratios and kinetic isotope effects

Sources	$\delta^{13}CH_4$ (‰)
AGR (agriculture)	Rice: -63[1,2] Ruminants: -67.9 (C-3)[1] and -54.5 (C-4)[1]
ENE (energy)	Coal: -41.2[1] Oil & gas: -44.6[1]
IND (industry)	-35[3]
TRA (road-transportation)	-26[3]
RCO (residential, commercial, and other)	-35[3]
WST (waste)	-55[2]
SHP (international shipping)	-40[4]
BMB (biomass burning)	C-3: -25[5] and C-4: -12[5]
OCN (ocean)	-58[2]
WET (wetland)	Spatially-resolved[6]
TMI (termite)	-57[7]
VOL (mud volcano)	-42.5[8]
Sinks	$\alpha^{13}C$ (kinetic $^{13}C/^{12}C$)
OH	0.9954 (1.0/1.00465) [9] (For EXP1-2, & 4) 0.9961 (1.0/1.0039)[9] (For EXP3)
O1D	0.9872 (1.0/1.013) [9]
Cl	0.9381 (1.0/1.066) [9]
Soil (dry deposition)	0.9823 (1.0/1.018)[10]

[1]Sherwood et al. (2017)
[2]Whiticar and Schaefer (2007)
[3]Averaged values based on Zazzeri et al. (2017) and Townsend-Small et al. (2015)
[4]Assumed based on the global mean isotopic ratios of fossil fuel
[5]Lassey et al. (2007)
[6]Ganesan et al. (2018)
[7]Houweling et al. (2000)
[8]Etiope et al. (2007)
[9]Saueressig et al. (2001)
[10]Snover and Quay (2000)

Table S3. Global annual mean observations for CH_4 and $\delta^{13}CH_4$

Year	CH_4 (ppb)[1]	$\delta^{13}CH_4$ (‰)[2]
1980	1585	-47.75
1981	1603	-47.70
1982	1619	-47.67
1983	1636	-47.62
1984	1645	-47.60
1985	1658	-47.55
1986	1671	-47.52
1987	1683	-47.50
1988	1693	-47.43

1989	1704	-47.40
1990	1715	-47.37
1991	1725	-47.40
1992	1735	-47.35
1993	1737	-47.37
1994	1742	-47.31
1995	1748	-47.27
1996	1751	-47.31
1997	1754	-47.23
1998	1765	-47.19
1999	1772	-47.20
2000	1773	-47.21
2001	1771	-47.19
2002	1773	-47.21
2003	1777	-47.19
2004	1777	-47.18
2005	1774	-47.22
2006	1774	-47.20
2007	1781	-47.21
2008	1787	-47.17
2009	1793	-47.25
2010	1798	-47.31
2011	1803	-47.35
2012	1808	-47.35
2013	1813	-47.37
2014	1823	-47.37
2015	1834	-47.40
2016	1843	-47.40
2017	1850	-47.40

[1] Values for 1980-1982 are based on the specified concentrations in support of climate modeling by Meinshausen et al. (2017) and values for 1983-2017 are based on NOAA GML surface observations (Lan et al., 2024). Numbers are rounded to the nearest integer.
[2] Values for 1980-2014 are from Schaefer et al. (2016) and values for 2015-2017 are scaled based on 2014 value, following the same trend of 2015-2017 calculated based on the measurements from the Institute of Arctic and Alpine Research (INSTAAR) (Michel et al., 2023). Numbers are rounded to the two digits. The uncertainties in the 2015-2017 values will not affect the overall results in this work.

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